

CareConnect — Explainable ED Triage Decision Support

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1. Problem statement

Emergency triage assigns an Emergency Severity Index (ESI 1–5) under time pressure, from one nurse's judgement, with no consistent second opinion and no record of why a level was chosen. Under-triage of ESI 1–2 patients is the costliest error. CareConnect adds a model that suggests an ESI level from the eleven values recorded at the first assessment, explains each suggestion, and requires a clinician to confirm or override it — decision support, never a decision.

2. Dataset description

Source. A reproducible synthetic ED generator (seed 42) for development, and MIMIC-IV-ED v2.2 (PhysioNet, Beth Israel Deaconess, 2011–2019) as real data: the open demo subset (207 triage rows, 64 patients) and the full credentialed release (~425,000 ED stays) for final evaluation.

Size and split. 12,000 synthetic rows, stratified 60/20/20 into 7,200 train / 2,400 validation / 2,400 test; patient-level grouping on real data so no patient spans splits.

Features. Eleven triage-time inputs — age, sex, arrival transport, heart rate, respiratory rate, systolic and diastolic BP, SpO₂, temperature, pain score, free-text chief complaint — plus two derived: shock index (HR/SBP) and pulse pressure (SBP–DBP). Target: ESI 1–5.

Preprocessing. Out-of-range values clipped to missing, median imputation and z-scoring for numerics, mode imputation and one-hot for categoricals, TF-IDF (1–2 grams, 400 terms, sublinear TF) over the complaint text. Everything stateful is fit on the training split only and serialized inside the model artifact. A named whitelist admits only triage-time columns; diagnosis, disposition, ICU admission, mortality, length of stay, later vitals, labs and notes are blocked by name.

3. Methodology

Pipeline: FeatureEngineer → ColumnTransformer → estimator. Four model families (logistic regression, decision tree, random forest, XGBoost) are each trained on two feature sets (structured, structured + TF-IDF), class-weighted for the imbalanced ESI distribution. Selection uses validation macro-F1 with ESI 1–2 recall as the tie-break — never accuracy; the winner is refit on train + validation and the test split is scored once. XGBoost (400 trees, depth 5, learning rate 0.05, multi:softprob) with TF-IDF is the expected selection. SHAP TreeExplainer returns the top five drivers per prediction, folded back to base features and labelled by direction of acuity, with a rule-based fallback that explains itself the same way. Every response carries requires_human_review, and the model output and the clinician's final ESI are stored side by side in Postgres so neither overwrites the other.

4. Results and insights (targets)

Metric	In-domain	Target A (demo)	Target B (full)
Accuracy	0.87	0.792	≥ 0.80
Macro-F1	0.85	0.536*	≥ 0.72
ESI 1–2 recall	0.93	0.913	≥ 0.93
ESI 1–2 precision	0.93	0.868	≥ 0.85
Calibration error (ECE)	≤ 0.03	—	≤ 0.03

* Target A is capped by its own cohort: the demo has no ESI 5 and two ESI 4 patients, so macro-F1 over five classes tops out at 0.80 even for a perfect model. Target B, on the full dataset, is the acceptance gate.



Dataset acceptance is tested statistically, not by eye: no vital significantly different from the reference cohort after Holm correction, all nine equivalent at a ± 0.2 SD TOST margin, χ^2 $p \geq 0.05$ on the ESI mix, and a mean absolute correlation gap ≤ 0.05 — including a pain–acuity correlation matching the reference in sign.

5. Novelty

- Multimodal triage prediction: structured vitals and free-text complaint in one pipeline, with the text contribution measured as a required gain rather than assumed.
- Explanations phrased in clinical direction — each SHAP driver is labelled as increasing or decreasing acuity against the ordinal ESI scale, not as a raw coefficient.
- Auditable human-in-the-loop: the model's prediction, probabilities, inputs and explanation are immutable; the clinician's final ESI, override flag and reason sit beside them.
- Privacy-safe calibration: the synthetic generator is tuned to real aggregate statistics only, with tests proving no patient row, identifier or complaint text can enter the generated data.
- A statistical acceptance gate on the training data itself — equivalence testing, not just a non-significant t-test, before the data is considered fit to train on.